ECO 202 — In-Class Exam 3

Statistics and Data Analysis for Economics | Princeton University

November 9, 2026 | Classes 10–13 | 100 points

Name: _____ Student ID: _____

Timing. Aim to complete your answers in 55–60 minutes, leaving 10–15 minutes to check within the 70-minute target. Follow announced start/end times within the 80-minute meeting.

Conditions. No books, notes, calculators, AI, or other electronics; University-authorized accommodations apply. Fractions, radicals, and equivalent exact expressions are acceptable. Show reasoning in written answers. Data are teaching examples unless identified otherwise.

Points. Questions 1–4: one answer A–D, 5 points each, no negative marking. Questions 5–8: 20 points each; subpart points are shown. Answer each part in its space; scratch pages 3 and 12 are not graded.

Contents and suggested pacing

Question	Topic	Page	Points	Minutes
1	Continuous distributions	2	5	2.5
2	Estimated standard error	2	5	2.5
3	Sample size and precision	2	5	2.5
4	Unbiasedness	2	5	2.5
5	Joint outcomes and combined risk	4	20	12
6	One sampling distribution, several criteria	6	20	12
7	Counts, proportions, and approximation	8	20	15
8	Calibrated measurements and estimation	10	20	10
Checking and review			—	11
Total			100	70

Scratch work: pages 3 and 12.

Princeton University Honor Code

At the end of the examination, write the pledge in full below and sign your name.

"I pledge my honor that I have not violated the Honor Code during this examination."

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Signature: _____

Multiple choice (20 points; about 10 minutes)

1. A continuous random variable T has density $f_T(t) = 1/2$ on $[0, 2]$ and zero elsewhere. Which statement is correct?

A. $\mathbb{P}(T = 1) = 1/2$.

B. $\mathbb{P}(1/2 \leq T \leq 3/2) = 1/2$.

C. $\mathbb{P}(T \leq 1) = 1$.

D. A density height is always a point probability.

2. An independent random sample of 25 observations from one population has sample variance $s^2 = 36$. What is the usual estimated standard error of its sample mean?

A. 6.

B. 6/5.

C. 36/25.

D. 6/25.

3. Under independent sampling from a fixed population with positive finite variance, increasing the sample size from 25 to 100 makes the sample mean's standard error:

A. One-half as large.

B. One-quarter as large.

C. Twice as large.

D. Unchanged.

4. An estimator T is unbiased for a fixed parameter θ. Which statement must hold?

A. Every realized estimate equals θ.

B. T has smaller MSE than every biased estimator.

C. The observations necessarily represent every population of interest.

D. $\mathbb{E}[T] = \theta$ under the stated sampling model.

Scratch work

Use this page for calculations. Work on this page is not graded. Written Questions 5–8 begin on the following page.

5. Joint outcomes and combined risk (20 points; about 12 minutes)

The entries below are joint probabilities for random variables X and Y.

	$X = 0$	$X = 2$
$Y = -1$	0	1/5
$Y = 1$	1/2	3/10

(a) (6 points) Find the marginal distribution of X. Then, calculate $\mathbb{E}[X]$ and $\text{Var}(X)$.

(b) (8 points) Calculate $\text{Cov}(X, Y)$ and $\text{Corr}(X, Y)$. You may use $\mathbb{E}[Y] = 3/5$ and $\text{Var}(Y) = 16/25$.

(c) (6 points) Find $\text{Var}(X + Y)$. Explain why omitting the covariance term would give an incorrect answer here.

6. One sampling distribution, several criteria (20 points; about 12 minutes)

A population of four individuals has recorded values $\{1, 3, 4, 5\}$. Select a simple random sample of three distinct individuals without replacement, so each three-individual subset is equally likely. Let T be the sample median. Consider T as an estimator of the *population mean* μ.

For this question you may use $\text{MSE}(T; \mu) = \text{Var}(T) + (\mathbb{E}[T] - \mu)^2$.

(a) (6 points) List the four possible samples and their medians. Give the sampling distribution of T.

(b) (4 points) Calculate μ, $\mathbb{E}[T]$, and the bias of T for μ.

(c) (5 points) Calculate $\text{Var}(T)$ and $\text{MSE}(T; \mu)$.

(d) (5 points) A student proposes using $\sigma^2/3$ as the variance of T, where σ^2 is the population variance of one uniformly chosen value. Give two reasons why this shortcut is not justified for this question.

7. Counts, proportions, and approximation (20 points; about 15 minutes)

In a teaching model, $H_1, \ldots, H_{100}$ are independent Bernoulli responses, each with success probability $p = 1/2$. Define the count $K = \sum_{i=1}^{100} H_i$ and the proportion $\widehat{P} = K/100$.

For a standard Normal variable Z, write $\Phi(z) = \mathbb{P}(Z \leq z)$. You may use $\Phi(1.90) \approx 0.9713$ and $\Phi(2.00) \approx 0.9772$.

(a) (6 points) Starting from the mean and variance of an indicator, derive the mean and variance of K and the mean and standard error of $\widehat{P}$. Identify where independence is used.

(b) (6 points) Use a continuity-corrected Normal approximation to calculate $\mathbb{P}(K \geq 60)$. Show the continuous boundary, standardized value, and probability. State why the answer is approximate rather than exact.

(c) (4 points) Extend the same model to n responses and define $\widehat{P}_n = (H_1 + \cdots + H_n)/n$. State what the Law of Large Numbers says about $\widehat{P}_n$ as n grows. Suppose $\widehat{P}_{100} = 0.6$. Does the LLN imply that $\widehat{P}_{101}$ must be closer to $1/2$? Explain.

(d) (4 points) Suppose instead that all 100 responses are copies of one Bernoulli draw H, so $H_i = H$ for every i. What is the actual standard error of $\widehat{P}$ in this case? Explain why increasing the number of identical copies does not improve precision.

8. Calibrated measurements and estimation (20 points; about 10 minutes)

Two independent measurements satisfy

$$\mathbb{E}[X] = \mu, \quad \mathrm{Var}(X) = 4, \qquad \mathbb{E}[Y] = 2\mu + 2, \quad \mathrm{Var}(Y) = 4.$$

The unknown target is μ. Define $A = (Y - 2)/2$. You may use $\mathrm{MSE}(T; \theta) = \mathrm{Var}(T) + (\mathbb{E}[T] - \theta)^2$.

(a) (8 points) Show that both X and A are unbiased for μ. Find both variances and determine which has smaller MSE.

(b) (6 points) Suppose μ is the mean wage of workers on a sampling list that excludes self-employed workers. An analyst makes two claims: (i) A is also unbiased for the mean wage of all workers, including the self-employed; (ii) more independent measurements from the same list would eliminate any mismatch with the population of all workers. Is each claim justified by the information given? Explain for each claim.

(c) (6 points) The observed value of Y is $y = 18$. Let a denote the corresponding realized value of A. Report a and the true standard error of A. Can these two numbers reveal the realized estimation error $a - \mu$? Explain why or why not.

Additional scratch work

Use this page for additional calculations. Work on this page is not graded.